

Urea-Based Ligand as an Efflux Pump Inhibitor: Warhead to Counter Ciprofloxacin Resistance and Inhibit Collagen Adhesion by MRSA

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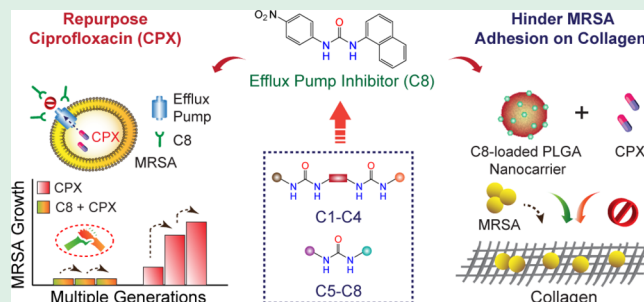
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ABSTRACT: Methicillin-resistant *Staphylococcus aureus* (MRSA) is a frontline human pathogen in which efflux pump activity confers high levels of antibiotic-resistance and poses a therapeutic challenge in the clinics. The present study illustrates the potential of urea-based ligand as an efflux pump inhibitor (EPI) in order to restore the efficacy of ciprofloxacin (CPX) against MRSA. Among eight structurally varying urea-based ligands, the ligand C8 could significantly inhibit efflux pump activity in the clinical MRSA strain *S. aureus* 4s and was superior to the known EPI reserpine. In combinatorial treatment, C8 enhanced cellular accumulation of CPX, rendered a 16× decrease in the MIC of CPX, and restored the susceptibility of *S. aureus* 4s to CPX. Notably, C8 downregulated the expression of *norA* gene coding for the efflux pump in MRSA and treatment with 10 μM C8 and 2.0 μM CPX prevented emergence of the CPX resistance trait and suppressed MRSA cell growth till 120 generations. For potential anti-MRSA therapy, C8-loaded poly(D,L-lactide-co-glycolide) nanocarrier (C8-PNC) was generated, which facilitated facile release of C8 in physiologically relevant fluid. C8-PNC (loaded with 50 μM C8) rendered efflux pump inhibition and eliminated MRSA in combination with only 2.0 μM CPX. Treatment with the non-toxic C8-PNC (loaded with 50 μM C8) and CPX (2.0 μM) also hindered MRSA adhesion on collagen manifold higher as compared to cells treated with 32 μM CPX and significantly downregulated *norA* gene expression in non-adhered MRSA cells. The urea-based ligand presented herein is a promising biocompatible therapeutic material for effective mitigation of MRSA infections.

KEYWORDS: MRSA, efflux pump inhibitor, resistance, ciprofloxacin, collagen



INTRODUCTION

Methicillin-resistant *Staphylococcus aureus* (MRSA) has emerged as a serious human pathogen as it is implicated in a large number of nosocomial and community-associated infections.^{1–3} MRSA is also known to be the causative agent of serious invasive infections in collagen rich tissues, wherein the initial adhesion onto the collagen matrix mediated by the adhesin protein present in the pathogen may play a critical role.^{1,4} Given the restricted number of antibiotics that are active against MRSA, development of an effective therapy against this pathogen is quite challenging. The current therapeutic interventions for severe MRSA infections largely encompass the use of antibiotics such as vancomycin, daptomycin, or linezolid.^{1,5} Additional antibiotics in the pipeline and under consideration as an anti-MRSA arsenal include ceftaroline, telavancin, delafloxacin, and others.¹ However, with reports emerging on resistance development against vancomycin,^{6,7} which is a frontline anti-MRSA therapeutic antibiotic, there is an urgent demand for an effective therapeutic strategy against the pathogen. To this end, combination therapy has emerged as a feasible choice as the synergy between two drugs can amplify the antibacterial effect, reinstate vulnerability of the target cells, and also prevent

development of drug-resistance during therapy.^{8–11} Combinatorial effect of gentamicin with therapeutic antibiotics like daptomycin and vancomycin has been reported for alleviation of *S. aureus*-mediated infection.^{12,13} Studies have also reported that combinatorial treatment of ceftioxin and β -lactams may hold interesting prospect against MRSA.¹⁴

It is anticipated that deployment of two different antibiotics in tandem would expand the activity spectrum and foster synergy between the antibiotics against MRSA. Further, such a combination can lower the effective therapeutic dose of the antibiotic for pathogen elimination. This premise is especially relevant for a treatment regime with aminoglycoside, fluoroquinolone, and glycopeptide-based antibiotics against MRSA, since these antibiotics are known to have toxic effects on the host.^{15–17} In this context, the use of a non-toxic synthetic molecule, which can target the core resistance

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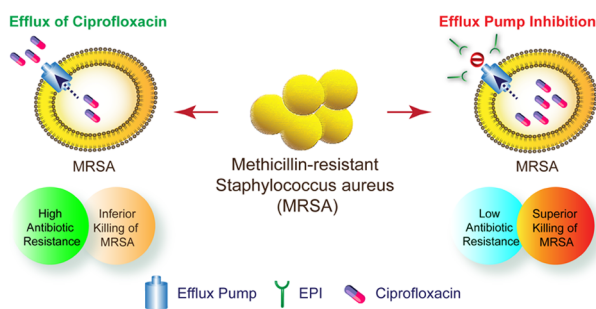
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mechanism of MRSA against a therapeutic antibiotic and thereby restore its susceptibility, offers an interesting prospect. Conceivably, the potential synergy rendered by the combination of such synthetic ligands and therapeutic antibiotics will also foster a dosing regimen that will hinder resistance development, enable effective use of the antibiotic to eliminate MRSA, and minimize host-directed toxic effects.

In staphylococci, the presence of efflux pumps render resistance against biocides, antiseptics, and therapeutic antibiotics.^{18–20} Hence, they can be considered as probable drug targets and discovery of an effective efflux pump inhibitor (EPI) remains a major endeavor in order to curtail MRSA infections. A key efflux pump in MRSA involved in the exclusion of the therapeutic fluoroquinolone ciprofloxacin (CPX) is the NorA protein.²⁰ A recent study has also highlighted that NorA is implicated in the development of CPX-resistance in *S. aureus*.²¹ Based on these reports, it can be surmised that monotherapy with CPX is likely to result in high resistance development and reduced killing of MRSA (Scheme 1). On the other hand, targeting MRSA with a synthetic EPI in

Scheme 1. Cartoon Illustrating the Implications of Engaging Synthetic Efflux Pump Inhibitor (EPI) in Combination Therapy for Curbing Ciprofloxacin Resistance and Achieve Superior Killing of MRSA



a combinatorial therapeutic regime can breach the core evasive mechanism through efflux pump inhibition, resulting in reduced resistance development and enhanced CPX-mediated elimination of MRSA (Scheme 1). This tenet has been vindicated in previous studies that highlight the prospect of synthetic EPIs in potentiating the bactericidal efficacy of CPX against *S. aureus*.^{22–26}

Previous studies have described a gamut of EPIs based on benzothiazine, quinoline, boronic, chalcone, and indole derivatives (Supporting Information, Figure S1).^{22–24,27–30} Structure–activity studies suggest that a medium molecular size, a noteworthy polar surface area, hydrophobicity and the presence of H-bond donor/acceptor groups are critical requisites that determine the activity of synthetic EPIs.^{23,30,31} The core design principle emerging from these studies can be leveraged as a guideline to generate synthetic EPIs that can counter CPX resistance in MRSA and restore activity of the antibiotic for potential clinical use. Further, for future clinical applications, design of non-bactericidal EPIs will be worthwhile as it would lower the possibility of resistance development against such candidates in the target pathogen. In this context, the current study illustrates the benefit of using urea-based ligands as EPI and demonstrates a structure-guided activity in the ligands. The study reports the effect of ligand C8, the most potent EPI, on the expression of *norA* gene coding for a prominent efflux pump in MRSA. The study illustrates that C8

could potentiate the activity of CPX and effectively hinder emergence of CPX-resistance in MRSA cells grown over multiple generations. For potential therapeutic applications, a non-toxic C8-loaded PLGA nanocarrier (C8-PNC) was generated, which restored susceptibility of MRSA to CPX and could effectively curb adhesion of MRSA cells onto the collagen matrix.

RESULTS AND DISCUSSION

Design Principle of Urea-Based Ligands. Previous structure–activity studies suggest that a medium molecular size, notable polar surface area, core hydrophobicity, and the presence of H-bond donor/acceptor groups are key features for design of a potent EPI.^{23,30,31} Based on this notion, in the current study, ligands bearing urea as the core building block were synthesized (Figure 1) by following previously reported

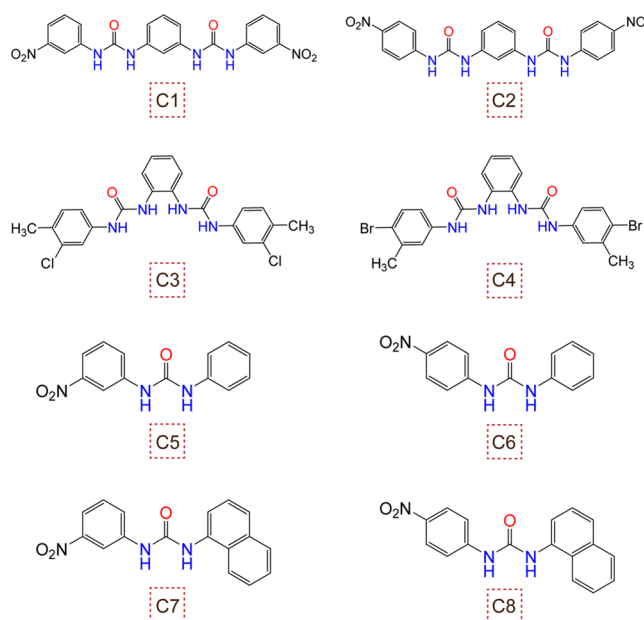


Figure 1. General structure of urea-based synthetic ligands (C1–C8).

methods.^{32–35} The central urea moiety can readily form H-bond assemblies and can also be functionalized easily to provide structural diversity. Ligands C1–C4 bear two urea units each and are likely to self-assemble more readily as compared to ligands C5–C8. Further, the two urea units render more hydrophilicity in C1–C4. Hence, it would be interesting to probe whether the presence of an additional urea unit in C1–C4 imparts any effect on efflux pump inhibition. It may be mentioned that C8 is more polar than C7 owing to the presence of the nitro group in the para position. Moreover, C8 is more hydrophobic than C6. Thus, selection of C5–C8 is likely to unravel the effect of hydrophobicity and polarity on efflux pump inhibition on a comparative basis.

Effect of Urea-Based Ligands on Efflux Pump Activity in MRSA Cells. Prior to assessing the potential of the urea-based ligands (C1–C8) as an EPI, their bactericidal activity against MRSA was determined. It was conceived that if the ligands are non-bactericidal, their merit in anti-MRSA therapy would be enhanced as the probability of resistance development against these agents will be reduced. To pursue this objective, cells of *S. aureus* 4s, which is a clinical MRSA strain, were grown in separate sets in the presence of the ligands

(5.0–80 μM). It was observed that none of the ligands displayed prominent bactericidal activity against MRSA at the tested concentrations (Figure S2). Further, for certain ligands such as C1, C2, C3, and C4, growth inhibition of MRSA was apparently higher at lower concentrations of the ligand (Figure S2). Two urea groups are present in these ligands as compared to a single urea group present in the ligands C5–C8 (Figure 1). Thus, it is presumed that ligands C1–C4 have a higher propensity to self-assemble in solution, especially at higher concentrations. This may in turn, hinder their antibacterial activity at higher concentrations.

Inhibition of efflux pump activity in the MRSA strain *S. aureus* 4s by the synthetic ligands was determined by the ethidium bromide (EtBr)-based efflux assay.³⁶ As per the well-established assay principle, it was conceived that efflux of EtBr will be favored in energized MRSA cells in the presence of glucose leading to a decrease in cell-associated EtBr fluorescence and inhibition of efflux activity by the ligands can be ascertained by retention of high cell-associated EtBr fluorescence (Figure 2A). To this end, MRSA cells exhibited

μM concentration (Figure S3 and Table S1). Quantitative estimation of efflux activity inhibition suggested that at an equimolar concentration of 10 μM , C8 could render the highest inhibition of EtBr efflux activity amounting to $\sim 95\%$ in the tested MRSA strain, followed by C1, C6, and C5 wherein the inhibitions of EtBr efflux activity were observed to be $\sim 66\%$, $\sim 46\%$ and $\sim 34\%$, respectively. The superior efflux pump inhibition rendered by C8 as compared to the other ligands was also evident when tested at other concentrations amounting to 5.0 and 40 μM , each (Figure S4). It may be noted here that the relative fluorescence intensity of EtBr was marginally less in the case of 40 μM C8 as compared to 5.0 μM of C8 (Figure S4). At 40 μM , the local concentration of C8 is likely to be less due to self-assembly in aqueous medium. This may account for the lower fluorescence intensity of EtBr when C8 was used at 40 μM in the efflux assay (Figure S4). Efflux pump inhibition by C1 and C8 was further substantiated by an EtBr accumulation assay, which revealed a strong proclivity of *S. aureus* 4s cells to accumulate EtBr in the presence of the ligands (Figure 2C). It may be mentioned that the tested concentration of C1 (40 μM) was manifold higher than that of C8 (10 μM). However, the highest cell-associated fluorescence was recorded for C8 (Figure 2C), reiterating that this ligand demonstrated the highest potency in terms of inhibition of efflux pump activity in MRSA. EtBr accumulation in C8-treated MRSA cells was significantly greater than that observed with the known EPI reserpine (Table S2). With regard to the cells subjected to treatment with the standard EPI reserpine in both EtBr efflux as well as the EtBr accumulation assay, it is anticipated that in an energized cell (cell subjected to prior incubation with glucose), which is predisposed to exhibit the active efflux of EtBr, treatment with reserpine should occlude the efflux of EtBr. Hence, in these cells, it is expected that the relative fluorescence intensity of EtBr will be comparable with that obtained for the control sample consisting of untreated cells (cells incubated without glucose). In this regard, it was observed that the relative EtBr fluorescence intensity values for untreated (control) and reserpine-treated samples were essentially comparable as the difference in fluorescence intensity between these samples (Figure 2C and Figures S2 and S3) were not significant based on a p value of <0.001 in ANOVA followed by an all-pairwise comparison (Holm–Sidak method). Based on the evaluation of the ligands, it is quite clear that C8 emerged as the most potent EPI. The ligands C1–C4 bear two urea units each, and the propensity to self-assemble is likely to be more in these ligands as compared to the ligands C5–C8. Self-assembly of the ligands C1–C4 may also hinder their interaction with the target efflux pump and reduce the efficacy of the ligands as EPI. In the case of C8, an electron-withdrawing nitro group, higher hydrophobicity as compared to C5 and C6 and a higher polarity as compared to C7 perhaps enhances its efficacy as an EPI, as corroborated by similar attributes present in other reported EPIs.^{23,27,31,30} The results emerging from the current study are interesting leads, which can perhaps be substantiated in future through more rigorous structure–function studies in order to derive guidelines for the design of potent EPIs.

Effect of C8 on Ciprofloxacin (CPX) Accumulation and Expression of *norA* Gene in MRSA. EtBr-based fluorescence assay clearly indicated that C8 could render strong inhibition of efflux pump activity in MRSA cells (Figure 2B–C). Since EtBr is a surrogate substrate for efflux pump implicated in export of fluoroquinolone such as CPX in

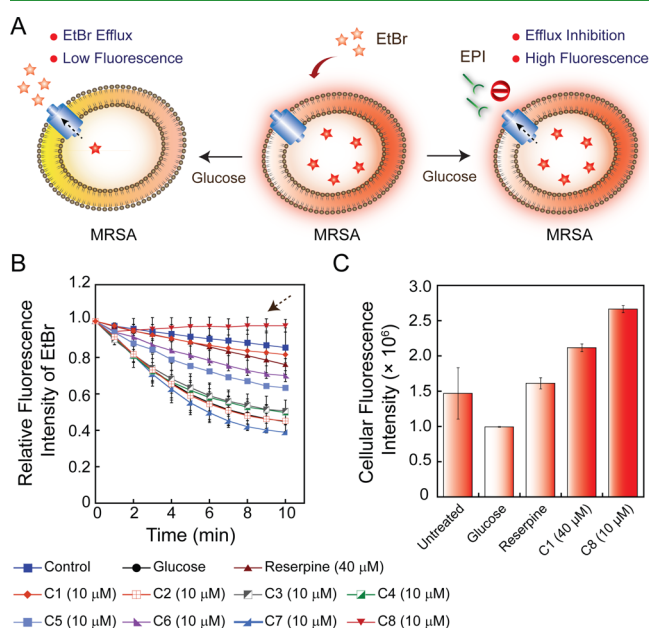


Figure 2. (A) Cartoon indicating the principle of the EtBr-based efflux pump assay and the effect of the efflux pump inhibitor (EPI). (B) Efflux assay in *S. aureus* 4s cells in the presence of urea-based ligands using EtBr as a substrate. The arrow indicates the trace obtained with C8. (C) Measurement of cell-associated EtBr fluorescence in *S. aureus* 4s cells treated with C1 and C8. Reserpine (40 μM) was used as positive control for both the assays.

active efflux of EtBr upon addition of glucose as the emission of the dye displayed a sharp decline over a period of 10 min (Figure 2B). Further, the lack of prominent EtBr efflux activity in the control sample (cells incubated without glucose) as well as in the sample with reserpine, a standard EPI, suggested the presence of efflux phenomena in the tested MRSA (Figure 2B). Among the ligands tested at an equimolar concentration of 10 μM , it was apparent that the ligands C1, C5, C6, and C8 seem to notably inhibit efflux pump activity in the MRSA strain (Figure 2B). Based on end-point analysis in the fluorescence-based assay, it was also evident that C8 (10 μM) exhibited superior efflux pump inhibition as compared to equivalent levels of other ligands and the known EPI reserpine used at 40

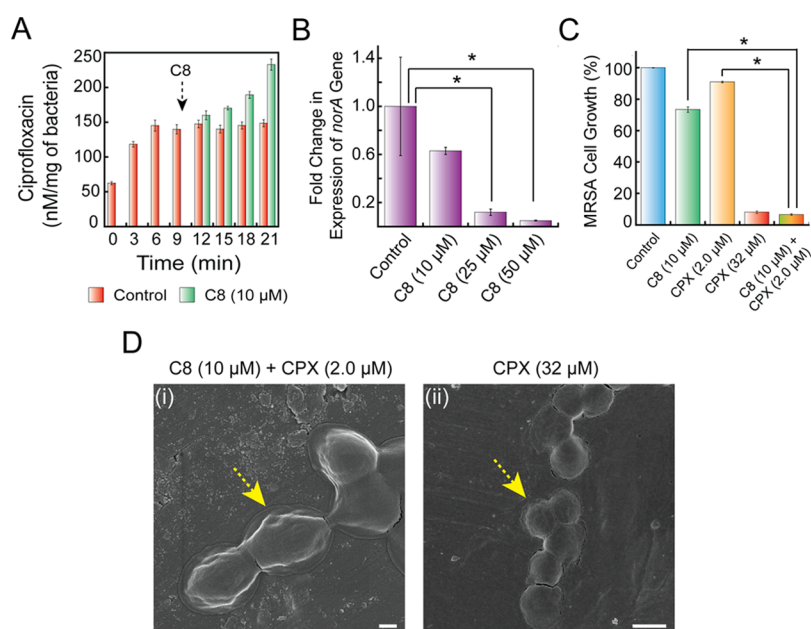


Figure 3. (A) Accretion of ciprofloxacin (CPX) in *S. aureus* 4s cells treated with 10 μ M C8. (B) Assessment of *norA* gene expression in *S. aureus* 4s treated with C8. (C) Effect of the combined treatment of C8 and CPX on the growth of *S. aureus* 4s. * in (B) and (C) indicates a *p* value of <0.001 in one-way ANOVA. (D) FESEM analysis to study the effect of (i) combination of C8 and CPX and (ii) CPX on *S. aureus* 4s cells. Yellow dotted arrows in the panels indicate distortion of typical morphology in treated cells. Scale bars in panels (i) and (ii) are 300 nm and 1.0 μ m, respectively.

MRSA,^{19,20} it was worthwhile to ascertain whether C8 could bolster CPX accumulation in MRSA cells. A standard fluorescence-based assay³⁷ indicated that in control cells (cells incubated with CPX alone), there was a gradual rise in the accrual of CPX, which reached a saturation level of $\sim$ 140 nM per mg of cell, following 9.0 min (Figure 3A). Interestingly, subsequent addition of 10 μ M C8 led to a conspicuous enhancement in the levels of CPX measured in MRSA cells. Notably, the final CPX level reached $\sim$ 232 nM per mg of cell after 21 min of assay time in C8-treated cells as against $\sim$ 148 nM per mg of cell in MRSA treated with CPX only (Figure 3A). Hence, C8 could impede the efflux process and enhance CPX accumulation in MRSA cells.

In clinical strains of MRSA, the NorA protein, which is a major facilitator superfamily (MFS) type of efflux pump^{18,20} is reported to be the causative membrane-associated protein for the expulsion of EtBr and CPX. In the *S. aureus* 4s strain, the recorded efflux pump activity (Figure 2B,C) suggested the presence of a NorA efflux pump. To this end, PCR revealed a *norA* gene amplicon having an expected size of $\sim$ 100 bp in *S. aureus* 4s strain (Figure S5). Although the presence of NorA, which is implicated in the efflux of EtBr and CPX²⁰ could be detected in the *S. aureus* 4s strain, the presence of other efflux pump proteins in this strain is also possible and this tenet can be validated in future studies.

Given that C8 could hinder efflux pump activity and augment CPX accumulation (Figures 2B,C and 3A), it was envisaged that C8 may alter the level of *norA* gene expression in the *S. aureus* 4s strain. Gene expression studies suggested a notable suppression of *norA* gene expression ($\sim$ 0.63-fold change) in MRSA cells treated with 10 μ M C8 as compared to the untreated control cells (Figure 3B). The decrease in *norA* gene expression in *S. aureus* 4s cells was even more significant in the presence of a higher concentration of C8, wherein the fold change in expression was $\sim$ 0.12 and $\sim$ 0.03 upon treatment with 25 and 50 μ M C8, respectively (Figure 3B).

In the context of anti-MRSA therapy, the remarkable downregulation of *norA* gene expression rendered by C8 augers well as a means to counter efflux pump activity and thereby curb CPX resistance.

Combinatorial Effect of C8 and Ciprofloxacin (CPX) on MRSA. The ligand C8 could not only render inhibition of efflux pump activity and enhance CPX accumulation but also downregulate *norA* gene expression in the MRSA strain *S. aureus* 4s (Figures 2B,C and 3A,B). Hence, it was pertinent to determine whether the bactericidal activity of CPX against *S. aureus* 4s can be restored in the presence of C8. Further, if the use of C8 in a combinatorial treatment regimen can reduce the effective bactericidal dose of CPX, it augers well as the antibiotic is known to render host-directed toxicity.¹⁶ The MIC of CPX against *S. aureus* 4s was 32 μ M (10.6 μ g/mL), which was in accordance with an earlier study.³⁸ In order to ascertain the adjuvant potential of C8 and determine the synergistic effect, a checkerboard assay was performed using varying concentrations of CPX (2.0–32 μ M) in combination with varying concentrations of C8 (5.0–40 μ M). Interestingly, the checkerboard assay revealed a decrease in the MIC of CPX in the presence of increasing concentrations of C8 (Table S3). For instance, in combination with 5.0, 20, and 40 μ M C8, the MIC of CPX was reduced to 4.0 μ M (Table S3). The observed synergy between C8 and CPX was highest in the case of a combinatorial treatment with 10 μ M of C8, wherein the MIC of CPX was 2.0 μ M and a 16-fold reduction in the MIC of CPX was observed (Table S3). In the checkerboard assay, there was a dose–response relationship in the presence of 5.0 and 10 μ M C8, wherein the MIC of CPX was reduced from 4.0 μ M in the presence of 5.0 μ M C8 to 2.0 μ M in the presence of 10 μ M C8 (Table S3). However, at higher concentrations of C8 (20 μ M and 40 μ M), the dose–response relationship was not observed as the MIC of CPX was enhanced to 4.0 μ M (Table S3). Self-assembly of C8 is likely to ensue at higher concentrations in aqueous medium, which may lead to a

reduction in the local concentration of the ligand. As a consequence of this phenomenon, the synergistic effect in the presence of higher concentrations of C8 (20 μ M and 40 μ M) was perhaps not as effective as observed in the case of 10 μ M C8, and a dose–response relationship was not realized. In this context, it can be mentioned that a similar phenomenon was also noted in our studies during the EtBr efflux assay, wherein the ability of C8 to inhibit EtBr efflux was reduced at higher concentrations, with the relative fluorescence intensity of EtBr being less in the case of 40 μ M C8 as compared to 5.0 μ M of C8 (Figure S4). It is also plausible that owing to the self-assembly of C8 at higher concentrations (20 and 40 μ M), CPX may interact with the assembly, which may result in a reduction in the effective concentration of CPX available for anti-MRSA activity in the combinatorial assay. Collectively, the aforementioned phenomenon likely accounts for the unusual trend of 2.0 μ M of ciprofloxacin exhibiting the highest inhibition activity against MRSA in the presence of 10 μ M of C8, but not at the higher concentrations of 20 or 40 μ M C8.

In this context of the combinatorial treatment regimen, it may be mentioned that C8 at a concentration of 10 μ M was essentially non-bactericidal but could inhibit efflux pump and enhance CPX accretion in *S. aureus* 4s cells (Figure S2 and Figures 2B,C and 3A). Growth inhibition of the target MRSA strain was only marginal in the presence of either 10 μ M C8 (~73% growth) or 2.0 μ M CPX (~91% growth) (Figure 3C). Interestingly, in the case of the combinatorial treatment with 10 μ M C8 and 2.0 μ M CPX, the growth of MRSA was significantly lower (~6.0% growth) than that observed in the presence of C8 or CPX alone (Figure 3C). It is also worthwhile to mention that the growth inhibition of MRSA rendered by 10 μ M C8 in combination with 2.0 μ M CPX was on par with 32 μ M CPX alone (~8.0% growth). Superior killing of MRSA cells in the presence of both C8 and CPX could also be observed in FESEM analysis. Herein, a significant perturbation of the typical morphology of *S. aureus* cells was evident, akin to treatment with 32 μ M CPX (Figure 3D). This was in contrast to the intact morphology observed in the case of untreated cells and cells treated with either C8 or CPX (Figure S6). Thus, it was evident that the potent efflux pump inhibition rendered by the ligand C8 could be leveraged in not only restoring the bactericidal activity of CPX but also mediating efficient elimination of MRSA using a very low concentration of CPX.

Potential of C8 in Curbing Ciprofloxacin (CPX) Resistance in MRSA. Manifestation of CPX resistance is unbridled in MRSA owing to the presence of efflux pump activity.^{18,20} In this context, the results obtained in the current study does seem to suggest that in a combinatorial treatment regimen, C8 holds interesting prospect to hinder efflux pump function and restore the susceptibility of MRSA to low levels of CPX. However, to validate the therapeutic merit of C8, it was pertinent to ascertain whether the ligand could also prevent the emergence of CPX-resistance in the MRSA strain *S. aureus* 4s upon prolonged exposure to the combinatorial effect of C8 and CPX. To investigate this aspect, the target cells were subjected to treatment in separate sets for 120 generations each with either 32 μ M CPX or a combination of 10 μ M C8 and 2.0 μ M CPX. Herein, it was observed that upon treatment with 32 μ M CPX alone (equivalent to its MIC against *S. aureus* 4s), growth of the target MRSA was curbed up to 40 generations (Figure 4A). However, with prolonged exposure to CPX, MRSA cells exhibited growth in the subsequent generations and a dramatic

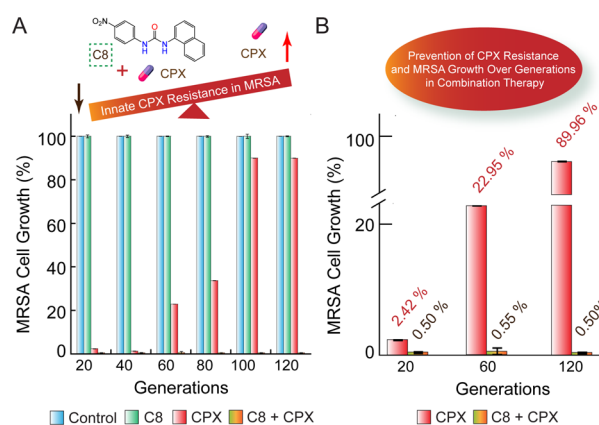


Figure 4. (A) Growth of *S. aureus* 4s cells cultured for multiple generations in the presence of CPX alone or a combination of C8 and CPX. (B) Expanded view of the plot in 4A indicating growth of *S. aureus* 4s cells cultured for multiple generations in the presence of either CPX alone or a combination of C8 and CPX.

revival was evidenced after 120 generations, with the cell growth reaching ~90% (Figure 4A,B). This suggested that CPX-resistance seems to evolve in the target MRSA with time, when the antibiotic was deployed alone, which was consistent with the presence of strong efflux pump activity detected earlier in the cells (Figure 2A). Interestingly, in the case of a combination treatment with 10 μ M C8 and 2.0 μ M CPX, growth of the target MRSA cells was completely subdued till 120 generations of growth (Figure 4A,B). This observation clearly indicated that the potent efflux pump inhibition rendered by C8 could be leveraged to effectively thwart emergence of ciprofloxacin resistance trait in MRSA cells. Thus, in a combinatorial treatment regimen, C8 holds interesting prospect as an adjuvant to contravene the inherent resistance mechanism and predispose MRSA to ciprofloxacin-mediated killing during a therapeutic window spanning multiple generations.

C8-Loaded PLGA Nanocarrier (C8-PNC). In order to explore the adjuvant potential of C8 in anti-MRSA therapy, it is necessary to develop a non-toxic delivery system for sustained release of the ligand. Based on the biocompatible and biodegradable attributes, PLGA nanoparticles (PNPs) were synthesized by a single emulsion method.³⁹ Microscopic analysis suggested that the synthesized PNPs were spherical in shape (Figure S7A–C). The average particle size of PNPs was ~200 nm as evident in FESEM analysis, and the average height profile assessed by AFM was ~225 nm (Figure S7D,E), while DLS indicated that the average hydrodynamic radius of PNP was ~396 nm (Figure S7F). In order to generate a C8-loaded PLGA nanocarrier (C8-PNC), PNPs were incubated with varying concentrations of C8. FESEM and AFM analysis indicated that C8-PNC was spherical in shape (Figure 5A–B), having a particle size of ~250 nm as determined by FESEM. The 3D topography AFM image revealed that the average height profile of C8-PNC was ~421 nm (Figure S8A,B), while the average hydrodynamic radius of C8-PNC as evident in DLS analysis was ~531 nm (Figure 5C). The presence of C8 in the nanocomposite was also ascertained by UV–visible absorbance spectroscopy, which indicated the characteristic absorbance peaks, typically associated with C8 (Figure S8C). The loading efficiency (LE) of PNP exhibited a systematic increase with increasing concentration of C8 and attained

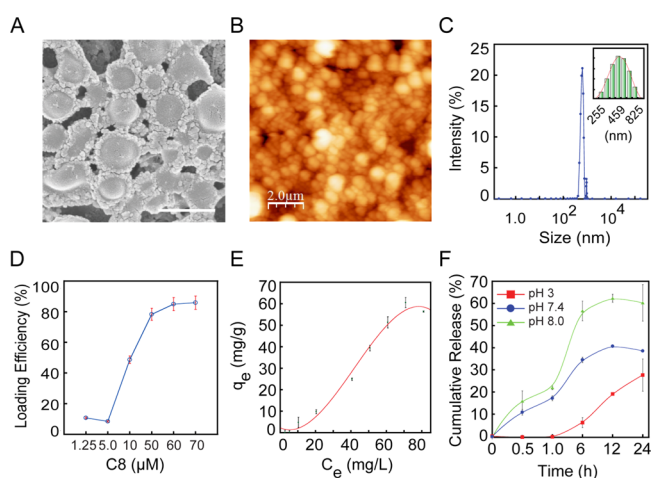


Figure 5. Characterization of the C8-loaded PLGA nanocarrier (C8-PNC) by (A) FESEM and (B) AFM analyses. The scale bar in (A) is 1.0 μm . (C) Estimation of the hydrodynamic radius of C8-PNC by DLS analysis. Inset indicates the size distribution histogram. (D) Loading efficiency of C8 onto the PLGA nanoparticle. (E) Adsorption isotherm of C8 on the PLGA nanoparticle. (F) Cumulative release kinetics of C8 from C8-PNC incubated in citrate buffer (pH 3.0), HEPES buffer (pH 7.4), and simulated body fluid (pH 8.0).

saturation at $\sim 60 \mu\text{M}$ of C8 (Figure 5D). LE was $\sim 86\%$ when the highest loading concentration of $70 \mu\text{M}$ C8 was tested (Figure 5D). Adsorption of C8 onto PNCs was consistent with a Langmuir isotherm model (Figure 5E) and suggested a homogeneous and monolayer.⁴⁶ Studies on release kinetics of C8 was performed with C8-PNC loaded with a high concentration of C8 ($80 \mu\text{M}$). Since the efficacy of an adjuvant molecule in a physiological milieu can be affected by its availability in serum and interactions with plasma protein and tissue, use of high concentrations of the adjuvant is warranted to ensure bioavailability and achieve the desired efficacy. At pH 3.0, the release of C8 was nominal and estimated to be $\sim 27\%$ following 24 h of incubation (Figure 5F). Interestingly, at higher pH of HEPES buffer and simulated body fluid (SBF), the release was sustained amounting to ~ 38 and $\sim 60\%$, respectively, after 24 h of incubation (Figure 5F). It is worthwhile to mention that the magnitudes of C8 released in the physiologically relevant HEPES buffer and SBF were ~ 30 and $\sim 48 \mu\text{M}$, respectively, which is manifold above the concentration required for effective NorA inhibition in the MRSA strain *S. aureus* 4s. Thus, the release profile of C8 from the nanocarrier augers well for potential therapeutic applications.

Efflux Inhibition and Adjuvant Activity of C8-PNC.

The EtBr-based fluorescence assay indicated a remarkable reduction in the efflux of EtBr in MRSA treated with C8-PNC, akin to cells treated with C8 alone (Figure 6A and Table S4). It may also be mentioned here that in the case of treatment with C8-PNC, an initial EtBr efflux was apparent till 2.0 min of assay time followed by a strong inhibition (Figure 6A). It is likely that following an initial lag period, cumulative release of C8 from the nanocarrier ensured that the ligand attained effective concentration in solution, which resulted in strong inhibition of efflux activity in MRSA cells. Based on the significant efflux inhibition rendered by C8-PNC, it was pertinent to ascertain whether C8-PNC could also enhance the efficacy of CPX in the combination assay and thereby hinder the growth of *S. aureus* 4s. A nominal growth inhibition of the

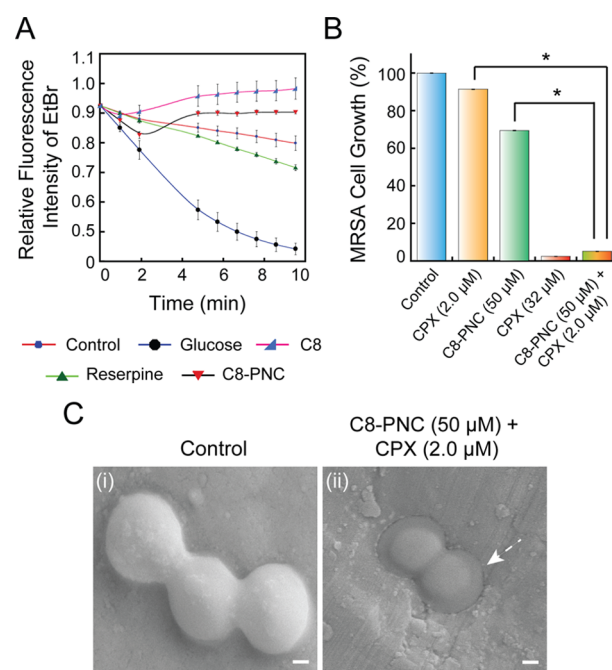


Figure 6. (A) EtBr efflux in *S. aureus* 4s cells treated with C8-PNC. Cells treated with reserpine ($40 \mu\text{M}$) constitute a positive control sample. (B) Growth of MRSA in the presence of C8-PNC (loaded with $50 \mu\text{M}$ C8) and ciprofloxacin ($2.0 \mu\text{M}$). * denoted p value < 0.001 in one-way ANOVA. (C) FESEM analysis to analyze the effect of treating *S. aureus* 4s cells with C8-PNC (loaded with $50 \mu\text{M}$ C8) and ciprofloxacin ($2.0 \mu\text{M}$). (i) Control sample consisting of untreated cells. The white arrow in panel (ii) indicates distortion of the typical morphology in treated cells. The scale bar is 200 nm.

MRSA strain was observed in the presence of either C8-PNC ($\sim 70\%$ growth) or $2.0 \mu\text{M}$ CPX ($\sim 91\%$ growth) (Figure 6B). However, combination treatment with C8-PNC (loaded with $50 \mu\text{M}$ C8) and CPX ($2.0 \mu\text{M}$) rendered a remarkable growth inhibition of *S. aureus* 4s cells, with the growth of MRSA cells being $\sim 7.0\%$ (Figure 6B). Notably, inhibition of MRSA cell growth rendered by C8-PNC in conjunction with $2.0 \mu\text{M}$ CPX was as efficient as that observed in the presence of $32 \mu\text{M}$ CPX (Figure 6B). It is also worth mentioning that the combinatorial treatment could reduce the MIC of CPX considerably ($16\times$ reduction) against *S. aureus* 4s. FESEM analysis provided supporting evidence of the effect of the combinatorial treatment, wherein significant distortion of the MRSA cell morphology was observed as compared to untreated control cells (Figure 6C). Thus, it was evident that the potent efflux pump inhibition rendered by the ligand C8 could also be leveraged in a nanocarrier-based delivery system, which not only supported a favorable release of the ligand and rendered inhibition of efflux activity in MRSA but was also effective in annihilation of MRSA when used in combination with a very low concentration of CPX.

Inhibition of MRSA Adhesion on Collagen by C8-PNC in Combination with CPX. Collagen adhesion is significantly implicated in initiating the infection process by *S. aureus*.^{1,4} The presence of collagen adhesin (Cna) in MRSA⁴ can favor the initial adhesion of planktonic cells onto collagen leading to biofilm formation, which can then pose a significant therapeutic challenge. To address this concern, a viable solution is to deter the initial adhesion of MRSA cells onto collagen. Encouraged by the potential of C8-PNC to inhibit

efflux activity and eliminate MRSA in combination with CPX, it was envisaged that this combinatorial treatment may hold prospect in inhibiting MRSA cell adhesion onto collagen. In the collagen adhesion assay, it was observed that untreated cells of MRSA displayed a very high propensity to adhere onto collagen ($\sim 95\%$ adhesion), while treatment with C8-PNC (loaded with $50\ \mu\text{M}$ C8) or CPX ($2.0\ \mu\text{M}$) resulted in only a marginal decrease in adhesion, amounting to ~ 87 and $\sim 88\%$ adhesion, respectively (Figure 7A). Notably, a combination

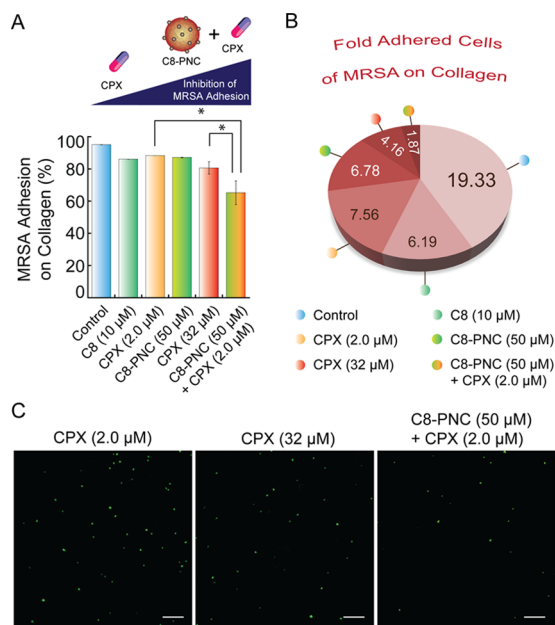


Figure 7. (A) Adhesion of *S. aureus* 4s cells onto collagen in the presence of various treatment regimens. * denotes a p value <0.001 in one-way ANOVA. (B) Quantification of fold adhered cells of MRSA in collagen adhesion assay conducted in the presence of various treatment regimens. (C) Fluorescence microscope analysis to visualize adhesion of MRSA cells onto collagen in the presence of various treatment regimens. The scale bar for the images is $20\ \mu\text{m}$.

treatment with C8-PNC (loaded with $50\ \mu\text{M}$ C8) and CPX ($2.0\ \mu\text{M}$) led to a substantial decrease in the magnitude of MRSA cell adhesion on collagen ($\sim 65\%$), which was significantly less than that observed in the case of treatment with either $2.0\ \mu\text{M}$ CPX or $32\ \mu\text{M}$ CPX (Figure 7A). This result was also verified by estimating the fold adhered cells (ratio of adhered and non-adhered MRSA cells), wherein the magnitude of cell adhesion on collagen was substantially lower in the case of combinatorial treatment (~ 1.87 fold) as compared to cells treated with either $2.0\ \mu\text{M}$ CPX (~ 7.56 fold) or $32\ \mu\text{M}$ CPX (~ 4.16 fold) (Figure 7B). A relative decrease in the population of adhered MRSA cells in the case of combinatorial treatment with C8-PNC (loaded with $50\ \mu\text{M}$ C8) and CPX ($2.0\ \mu\text{M}$) as compared to other treatment regimen was also evident in fluorescence microscopy (Figure 7C and Figure S9), and these results were also consistent with the trend observed in the quantitative adhesion assay (Figure 7A). It was thus apparent that treatment of MRSA cells with CPX alone during the assay may not deter cell adhesion onto collagen, although at a higher concentration equal to MIC of CPX against *S. aureus* 4s ($32\ \mu\text{M}$ CPX) the antibiotic is growth inhibitory. On the other hand, inclusion of C8-PNC (loaded with $50\ \mu\text{M}$ C8) in the combinatorial treatment could

effectively hinder CPX efflux and thereby potentiate the action of CPX on MRSA cells during the entire duration of the adhesion assay. This perhaps resulted in a decrease in the propensity of the treated MRSA cells to adhere onto collagen.

Effect of C8-PNC and CPX on *norA* Gene Expression in MRSA during Collagen Adhesion and Cytotoxic Potential of C8-PNC. The inhibition of *norA* gene expression in MRSA cells by C8-PNC and CPX during adhesion onto collagen was probed by real-time PCR analysis of non-adhered planktonic MRSA cells recovered after 1 h incubation from sets treated with C8-PNC (loaded with $50\ \mu\text{M}$ C8) and $2.0\ \mu\text{M}$ CPX. Treatment of the MRSA strain *S. aureus* 4s with $2.0\ \mu\text{M}$ CPX alone during collagen adhesion assay triggered a prominent upregulation of *norA* gene expression (~ 4.0 -fold) in non-adhered planktonic cells (Figure 8A). In MRSA, the

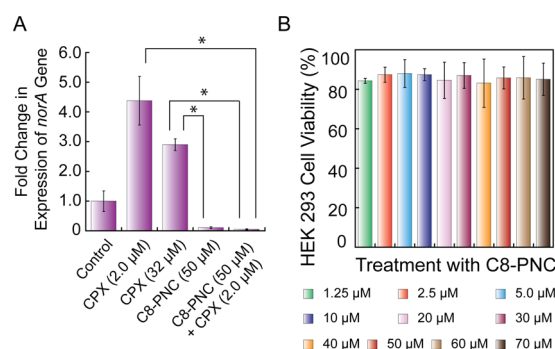


Figure 8. (A) Assessment of *norA* gene expression in non-adhered cells of *S. aureus* 4s recovered from the collagen adhesion assay. * denotes a p value of <0.001 in one-way ANOVA. (B) Cytotoxic effect of C8-PNC on HEK 293 cells determined by the MTT assay. The concentration of C8 loaded in C8-PNC is indicated. Estimation of mean $\pm$ standard deviation for each data point was accomplished from six experimental samples.

efflux pump NorA is implicated in the CPX efflux.^{18,20} In the light of this premise, exposure of MRSA cells to $2.0\ \mu\text{M}$ CPX likely induced active turnover of *norA* gene transcript. In the presence of $32\ \mu\text{M}$ CPX, equivalent to the MIC of the antibiotic against the target MRSA, *norA* gene expression in non-adhered planktonic cells was still upregulated (~ 3.0 -fold), albeit less in magnitude as compared to $2.0\ \mu\text{M}$ CPX (Figure 8A). It remains to be ascertained whether a higher concentration of CPX ($32\ \mu\text{M}$) imparts additional physiological perturbations in MRSA cells, which may lead to a decrease in *norA* gene transcript levels as compared to cells treated with only $2.0\ \mu\text{M}$ CPX. Interestingly, treatment of MRSA cells with C8-PNC (loaded with $50\ \mu\text{M}$ C8) either alone or in combination with $2.0\ \mu\text{M}$ CPX resulted in a dramatic downregulation in *norA* gene expression levels in non-adhered planktonic cells as compared to the untreated MRSA cells (Figure 8A). Based on these results, it is evident that C8-PNC can suppress *norA* gene expression and deter efflux pump activity in planktonic MRSA cells subjected to collagen adhesion and thus hold considerable potential as an anti-MRSA arsenal. In order to leverage C8-PNC in anti-MRSA therapy, it is critical that the nanocarrier should be non-toxic to host cells. An MTT-based assay indicated that C8-PNC did not impart any cytotoxic effect on human embryonic kidney cells (HEK 293 cells), wherein a cell viability in excess of 80% was observed even at a ligand loading concentration of $50\ \mu\text{M}$ C8 (Figure 8B), which rendered effective efflux

inhibition and elimination of MRSA in combination with 2.0 μM CPX (Figure 6A,B). Notably, C8 alone was indeed quite toxic to HEK 293 cells (Table S5). For instance, at a concentration of 10 μM , C8 could render significant inhibition of efflux activity in MRSA (Figure 2B). However, at this concentration, C8 was also toxic and suppressed the viability of HEK 293 cells ($\sim 42\%$ viability) (Table S5). In the case of treatment with C8 alone, the concentration of the ligand near HEK 293 cells is likely high, which may trigger a cytotoxic effect. On the other hand, in the case of C8-PNC, controlled release of C8 from the nanocarrier can reduce the local concentration of the ligand and thereby prevent manifestation of a toxic effect.

CONCLUSIONS

The prevalence of a rapidly evolving drug-resistance trait in MRSA in conjunction with the life-threatening infections caused by the pathogen is a daunting healthcare challenge. In MRSA, efflux pumps contribute significantly to resistance against therapeutic antibiotics and hence targeting the efflux pump can be considered as a viable therapeutic option. The current study is an endeavor in this direction, wherein a rationally designed urea-based ligand (C8) rendered effective inhibition of efflux pump and restored susceptibility of a clinical MRSA strain to low concentrations of CPX. The leads emerging from the study are interesting, which can perhaps be advanced in the future through rigorous structure–function studies in order to derive design guidelines for potent EPs. The ligand C8 could not only counter efflux pump activity in MRSA but could also prevent emergence of CPX resistance in the pathogen over multiple generations in combination therapy. Our study also demonstrates that the C8-loaded PLGA nanocarrier (C8-PNC) could be leveraged as an adjuvant as the nanomaterial was non-toxic, facilitated favorable release of the cargo in physiological fluid, and rendered manifold reduction in the dose of CPX for effective annihilation of MRSA. Interestingly, the developed nanomaterial could also curtail adhesion of MRSA cells onto collagen in combination with CPX and thus holds significant prospect in mitigation of MRSA-based invasive soft tissue infections. Given the limitation of antibiotic monotherapy against MRSA, the urea-based ligand described in the current study can serve as a prototype therapeutic material that can counter resistance and reinforce antibiotic-mediated anti-MRSA therapy.

EXPERIMENTAL SECTION

Bacterial Strain and Growth Conditions. *Staphylococcus aureus* 4s, a clinical MRSA strain was used in the current study. The target pathogen was propagated in BHI broth at 37 $^{\circ}\text{C}$ and 180 rpm for 12 h.

Urea-Based Ligands. The urea-based ligands (C1–C8) were synthesized by following the methods reported in earlier studies.^{32–35} Stock solution for each ligand (10 mg/mL) was prepared in DMSO, and the required working concentration of the ligands for each experiment was prepared accordingly from the stocks.

Anti-MRSA Activity of Urea-Based Ligands and CPX. The antibacterial activity of the urea-based ligands and CPX against *S. aureus* 4s was assessed by a broth dilution method as mentioned in the Supporting Information.

Efflux Pump Inhibition Assay. A fluorescence-based efflux pump inhibition assay was performed with *S. aureus* 4s cells by essentially following a previously reported method.³⁸ In separate sets, the assay was conducted with MRSA cells incubated with EtBr (5.0 $\mu\text{g}/\text{mL}$) for 1 h at 37 $^{\circ}\text{C}$ followed by treatment with ligands C1–C8 (5.0, 10, and

40 μM each) or C8-PNC (1.0 mg/mL of PNP loaded with 50 μM C8) or PNP (1.0 mg/mL) or the known efflux pump inhibitor reserpine (40 μM). A one-way analysis of variance (ANOVA) was performed for relative end-point fluorescence obtained in the EtBr efflux assay. Further, based on relative end-point fluorescence, the efflux activity in the *S. aureus* 4s strain was expressed as a percent fluorescence decrease over a 10 min time period and a quantitative estimation of EtBr efflux inhibition (%) was accomplished by following a previously described method.³⁰ Accumulation of EtBr in MRSA cells treated in separate sets with ligands C1–C8 (5.0, 10, and 40 μM each), C8-PNC (1.0 mg/mL of PNP loaded with 50 μM C8), PNP (1.0 mg/mL), or reserpine (40 μM) was determined as mentioned in the Supporting Information.

CPX Accumulation Assay. Accumulation of CPX in *S. aureus* 4s cells treated with C8 (10 μM) was measured by a standard method.³⁸

Effect of C8 on *norA* Gene Expression in MRSA. *S. aureus* 4s cells ($\sim 10^6$ CFU/mL) were grown in separate sets in BHI media supplemented with C8 (10, 25, and 50 μM) at 37 $^{\circ}\text{C}$ and 180 rpm for 24 h. Subsequently, the total RNA was isolated from the cells and 200 ng of RNA from each sample was subjected to quantitative real-time PCR. The gene specific primers used in PCR are indicated in Table S6. RNA isolation from *S. aureus* 4s and real-time PCR analysis was accomplished by following a previous method.⁴¹ The fold change in *norA* gene expression was determined by the $\Delta\Delta C_T$ method.⁴² Statistical analysis for fold change in *norA* gene expression was performed by a one-way analysis of variance (ANOVA).

Adjuvant Activity of C8. A 10 μL aliquot of *S. aureus* 4s cell suspension ($\sim 10^6$ CFU suspended in BHI) was inoculated into separate wells of a sterile 96-well microtitre plate containing 100 μL BHI medium supplemented with CPX (2.0–32 μM). In the case of combination treatment sets, a checkerboard assay was performed wherein varying concentrations of C8 (5.0, 10, 20, and 40 μM) was used in separate sets with every chosen concentration of CPX (2.0–32 μM). The treatment of cells was accomplished at 37 $^{\circ}\text{C}$ and 180 rpm for 12 h. Cell growth was ascertained by measuring absorbance at 600 nm (Infinite M200, TECAN, Switzerland) and expressed as percentage growth compared to untreated cells. In separate experiments, the effect of varying concentrations of CPX or C8 alone on the growth of target bacteria was also ascertained. The magnitude of decrease in MIC of CPX in the presence of C8 was determined. Statistical analysis for MRSA cell growth upon combinatorial treatment (10 μM C8 and 2.0 μM CPX) as compared to cells treated with either 10 μM C8 or 2.0 μM CPX was performed by a one-way analysis of variance (ANOVA) using Sigma Plot version 11.0. The adjuvant potential of C8 was also ascertained by FESEM analysis as described in the Supporting Information.

In Vitro Resistance of MRSA against CPX in the Presence of C8 over Multiple Generations. A 30 μL aliquot of overnight grown *S. aureus* 4s cell suspension was added into test tubes containing 3.0 mL BHI media supplemented in separate sets with (a) 10 μM C8 only, (b) 32 μM CPX only, and (c) combination of 10 μM C8 and 2.0 μM CPX. The tubes were kept in a shaker incubator for 12 h at 37 $^{\circ}\text{C}$ and 180 rpm. Following incubation, MRSA cell growth was measured by recording the absorbance at 600 nm (Lambda 25, Perkin-Elmer) followed by determining the percentage growth compared to untreated cells. A 30 μL aliquot of *S. aureus* 4s cell suspension from the respective treatment sets was used as an inoculum and again added into a separate test tubes and subjected to an identical treatment regimen. This cycle was repeated for a total of seven independent steps or 120 generations of *S. aureus* 4s cell growth. The number of generations was determined from the doubling time (t_d) of *S. aureus* 4s in BHI medium and total time of cell growth for each treatment cycle. The protocol for determination of the doubling time (t_d) of *S. aureus* 4s in BHI medium is described in the Supporting Information.

Generation of Nanocarrier (C8-PNC). PLGA nanoparticle (PNP) was prepared by a method as described earlier.³⁹ PNPs (1.0 mg/mL in sterile MilliQ water) were incubated in separate sets overnight under a rocking condition at room temperature with C8 (1.25–100 μM) for the generation of C8-loaded nanocarrier (C8-

PNC). The nanocarrier was then recovered by centrifugation at 10,000 rpm for 5.0 min and stored in -20°C for further use. Characterization of PNCs and C8-PNC was pursued by standard microscopic and spectroscopic methods as described in the [Supporting Information](#).

Loading Efficiency (LE), Adsorption Isotherm, and *In Vitro* Release Kinetics Studies. Experiments were conducted with C8-PNC to ascertain the loading efficiency (LE), adsorption isotherm, and cumulative release profile. The experimental procedures are mentioned in the [Supporting Information](#).

Adjuvant Activity of C8-PNC. The details of the experiments conducted to assess the adjuvant activity of C8-PNC is described in the [Supporting Information](#).

Effect of C8 and C8-PNC on Adhesion of MRSA onto Collagen. The collagen adhesion assay was performed by essentially following a previously described method.⁴³ For the adhesion assay, a 6-well tissue culture plate was coated with collagen solution overnight at 4°C (final concentration of $500\ \mu\text{g/mL}$). Washing and blocking of the wells was performed as reported earlier.⁴⁴ For the collagen adhesion assay, *S. aureus* 4s cells were labeled by cFDA-SE.⁴⁵ A 1.0 mL aliquot of cFDA-SE-labeled MRSA was then added to collagen-coated wells and incubated for 1 h at 4°C in separate sets with (a) C8 ($10\ \mu\text{M}$), (b) CPX ($2.0\ \mu\text{M}$), (c) C8-PNC (loaded with $50\ \mu\text{M}$ C8), (d) CPX ($32\ \mu\text{M}$), and (e) combination of CPX ($2.0\ \mu\text{M}$) and C8-PNC (loaded with $50\ \mu\text{M}$ C8). In a parallel set, untreated MRSA cells were also included. The solution fluorescence of cFDA-SE labeled cells measured at 518 nm upon excitation at 488 nm was considered as total fluorescence (F_T). Following incubation, non-adhered cells were aspirated and their fluorescence intensity was measured (F_{NA}). A quantitative measure of MRSA adhesion (F_A) onto collagen was obtained by estimating the difference between total fluorescence (F_T) and fluorescence intensity for non-adhered cells (F_{NA}). To study the expression of *norA* gene in *S. aureus* 4s cells subjected to various treatments, RNA extraction from non-adhered planktonic cells recovered in the adhesion assay experiment was performed followed by qRT-PCR analysis to ascertain fold change in the expression of *norA* gene as mentioned before. Fluorescence microscope analysis (Eclipse Ti-U, Nikon, USA) was also pursued to detect collagen-adhered MRSA cells.

Cytotoxicity Assay. The cytotoxic effect of C8 (5.0 – $40\ \mu\text{M}$) and C8-PNC (loaded with 1.25 – $70\ \mu\text{M}$ of C8) was ascertained in separate sets against the cultured human embryonic kidney (HEK 293) cell line by a standard MTT-based assay. Propagation of HEK 293 cells for the MTT assay and interpretation of the assay was based on a previously described method.³⁸

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsabm.2c00092>.

Additional experimental procedures; antibacterial activity of urea-based synthetic ligands (C1–C8) against the MRSA; effect of urea-based ligands on EtBr efflux in MRSA; checkerboard assay; PCR for *norA* gene detection in MRSA; FESEM analysis for C8- and CPX-treated MRSA; characterization of PLGA nanoparticle; AFM image and UV–visible absorption spectroscopy of C8-PNC; fluorescence microscope analysis to visualize adhesion of MRSA onto collagen; and cytotoxic effect of C8 on HEK 293 cells ([PDF](#))

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Notes

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■ REFERENCES

- (1) Lee, A. S.; De Lencastre, H.; Garau, J.; Kluytmans, J.; Malhotra-kumar, S.; Peschel, A.; Harbarth, S. Methicillin-resistant *Staphylococcus aureus*. *Nat. Rev. Dis. Prim.* **2018**, *4*, 18033.
- (2) Turner, N. A.; Sharma-Kuinkel, B. K.; Maskarinec, S. A.; Eichenberger, E. M.; Shah, P. P.; Carugati, M.; Holland, T. L.; Fowler, V. G., Jr. Methicillin-resistant *Staphylococcus aureus*: an overview of basic and clinical research. *Nat. Rev. Microbiol.* **2019**, *17*, 203–218.
- (3) Chambers, H.; Deleo, F. Waves of resistance: *Staphylococcus aureus* in the Antibiotic Era. *Nat. Rev. Microbiol.* **2009**, *7*, 629–641.
- (4) Foster, T. J.; Geoghegan, J. A.; Ganesh, J. A.; Höök, M. Adhesion, invasion and evasion: the many functions of the surface proteins of *Staphylococcus aureus*. *Nat. Rev. Microbiol.* **2014**, *12*, 49–62.
- (5) Liu, C.; Bayer, A.; Cosgrove, S. E.; Daum, R. S.; Fridkin, S. K.; Gorwitz, R. J.; Kaplan, S. L.; Karchmer, A. W.; Levine, D. P.; Murray, B. E.; Rybak, M. J.; Talan, D. A.; Chambers, H. F. Clinical practice guidelines by the Infectious Diseases Society of America for the treatment of methicillin-resistant *Staphylococcus aureus* infections in adults and children. *Clin. Infect. Dis.* **2011**, *52*, e18–e55.
- (6) Weigel, L. M.; Clewell, D. B.; Gill, S. R.; Clark, N. C.; McDougal, L. K.; Flannagan, S. E.; Kolonay, J. F.; Shetty, J.; Killgore, G. E.; Tenover, F. C. Genetic analysis of a high-level vancomycin-resistant isolate of *Staphylococcus aureus*. *Science* **2003**, *302*, 1569–1571.
- (7) Rossi, F.; Diaz, L.; Wollam, A.; Panesso, D.; Zhou, Y.; Rincon, S.; Narechania, A.; Xing, G.; Di Gioia, T. S. R.; Doi, A.; Tran, T.; Reyes, J.; Munita, J. M.; Carvajal, L. P.; Hernandez-Roldan, A.; Brandão, D.; van der Heijden, I. M.; Murray, B. E.; Planet, P. J.; Weinstock, G. M.; Arias, C. A. Transferable vancomycin resistance in

- a community-associated MRSA lineage. *N. Engl. J. Med.* **2014**, *370*, 1524–1531.
- (8) Fischbach, M. Combination therapies for combating antimicrobial resistance. *Curr. Opin. Microbiol.* **2011**, *14*, S19–S23.
- (9) Worthington, R. J.; Melander, C. Combination approaches to combat multidrug-resistant bacteria. *Trends Biotechnol.* **2013**, *31*, 177–184.
- (10) Thangamani, S.; Mohammad, H.; Abushahba, M. F. N.; Sobreira, T. J. P.; Hedrick, V. E.; Paul, L. N.; Seleem, M. N. Antibacterial activity and mechanism of action of auranofin against multi-drug resistant bacterial pathogens. *Sci. Rep.* **2016**, *6*, 22571.
- (11) Davis, J.; Van Hal, S.; Tong, S. Combination antibiotic treatment of serious methicillin-resistant *Staphylococcus aureus* infections. *Semin. Respir. Crit. Care Med.* **2015**, *36*, 3–16.
- (12) Tsuji, B. T.; Rybak, M. J. Short-course gentamicin in combination with daptomycin or vancomycin against *Staphylococcus aureus* in an *in vitro* pharmacodynamic model with simulated endocardial vegetations. *Antimicrob. Agents Chemother.* **2005**, *49*, 2735–2745.
- (13) Houlihan, H. H.; Mercier, R. C.; Rybak, M. J. Pharmacodynamics of vancomycin alone and in combination with gentamicin at various dosing intervals against methicillin-resistant *Staphylococcus aureus*-infected fibrin-platelet clots in an *in vitro* infection model. *Antimicrob. Agents Chemother.* **1997**, *41*, 2497–2501.
- (14) Banerjee, R.; Fernandez, M. G.; Enthaler, N.; Graml, C.; Greenwood-Quaintance, K. E.; Patel, R. Combinations of cefoxitin plus other β -lactams are synergistic *in vitro* against community associated methicillin-resistant *Staphylococcus aureus*. *Eur. J. Clin. Microbiol. Infect. Dis.* **2013**, *32*, 827–833.
- (15) Cosgrove, S. E.; Vigliani, G. A.; Campion, M.; Fowler, V. G.; Abrutyn, E.; Corey, G. R.; Levine, D. P.; Rupp, M. E.; Chambers, H. F.; Karchmer, A. W.; Boucher, H. W. Initial low-dose gentamicin for *Staphylococcus aureus* bacteremia and endocarditis is nephrotoxic. *Clin. Infect. Dis.* **2009**, *48*, 713–721.
- (16) Owens, R. C., Jr.; Ambrose, P. G. Antimicrobial safety: focus on fluoroquinolones. *Clin. Infect. Dis.* **2005**, *41*, S144–S157.
- (17) Finch, R. G.; Eliopoulos, G. M. Safety and efficacy of glycopeptide antibiotics. *J. Antimicrob. Chemother.* **2005**, *55*, ii5–ii13.
- (18) Li, X. Z.; Nikaido, H. Efflux-mediated drug resistance in bacteria: an update. *Drugs* **2009**, *69*, 1555–1623.
- (19) Costa, S. S.; Viveiros, M.; Rosato, A. E.; Melo-Cristino, J.; Couto, I. Impact of efflux in the development of multidrug resistance phenotypes in *Staphylococcus aureus*. *BMC Microbiol.* **2015**, *15*, 232.
- (20) Jang, S. Multidrug efflux pumps in *Staphylococcus aureus* and their clinical implications. *J. Microbiol.* **2016**, *54*, 1–8.
- (21) Papkou, A.; Hedge, J.; Kapel, N.; Young, B.; MacLean, R. C. Efflux pump activity potentiates the evolution of antibiotic resistance across *S. aureus* isolates. *Nat. Commun.* **2020**, *11*, 3970.
- (22) Sabatini, S.; Kaatz, G. W.; Rossolini, G. M.; Brandini, D.; Fravalini, A. From phenothiazine to 3-phenyl-1,4-benzothiazine derivatives as inhibitors of the *Staphylococcus aureus* NorA multidrug efflux pump. *J. Med. Chem.* **2008**, *51*, 4321–4330.
- (23) Sabatini, S.; Gosetto, F.; Manfroni, G.; Tabarrini, O.; Kaatz, G. W.; Patel, D.; Cecchetti, V. Evolution from a natural flavones nucleus to obtain 2-(4 propoxyphenyl) quinoline derivatives as potent inhibitors of the *S. aureus* NorA efflux pump. *J. Med. Chem.* **2011**, *54*, S722–S736.
- (24) Sabatini, S.; Gosetto, F.; Iraci, N.; Barreca, M. L.; Massari, S.; Sancineto, L.; Manfroni, G.; Tabarrini, O.; Dimovska, M.; Kaatz, G. W.; Cecchetti, W. Re-evolution of the 2-phenylquinolines: ligand-based design, synthesis, and biological evaluation of a potent new class of *Staphylococcus aureus* NorA efflux pump inhibitors to combat antimicrobial resistance. *J. Med. Chem.* **2013**, *56*, 4975–4989.
- (25) Radix, S.; Jordheim, A. D.; Rocheblave, L.; N'Digo, S.; Prignon, A.-L.; Commun, C.; Michalec, S.; Dijoux-Franca, M.-G.; Mularoni, A.; Walchshofer, N. N'-disubstituted cinnamide derivatives potentiate ciprofloxacin activity against overexpressing NorA efflux pump *Staphylococcus aureus* 1199B strains. *Eur. J. Med. Chem.* **2018**, *150*, 900–907.
- (26) Sundaramoorthy, N. S.; Mitra, K.; Ganesh, J. S.; Makala, H.; Latha, R.; Bhanuvalli, S. R.; Ulaganathan, V.; Tiru, V.; Sivasubramanian, A.; Nagarajan, S. Ferulic acid derivative inhibits NorA efflux and in combination with ciprofloxacin curtails growth of MRSA *in vitro* and *in vivo*. *Microb. Path.* **2018**, *124*, 54–62.
- (27) Sabatini, S.; Gosetto, F.; Serritella, S.; Manfroni, G.; Tabarrini, O.; Iraci, N.; Brincat, J. P.; Carosati, E.; Villarini, M.; Kaatz, G. W.; Cecchetti, V. Pyrazolo[4,3-c][1,2]benzothiazines 5,5-dioxide: a promising new class of *Staphylococcus aureus* NorA efflux pump inhibitors. *J. Med. Chem.* **2012**, *55*, 3568–3572.
- (28) Fontaine, F.; Hequet, A.; Voisin-Chiret, A.-S.; Bouillon, A.; Lesnard, A.; Cresteil, T.; Jolival, C.; Rault, S. First identification of boronic species as novel potential inhibitors of the *Staphylococcus aureus* NorA efflux pump. *J. Med. Chem.* **2014**, *57*, 2536–2548.
- (29) Gaur, R.; Gupta, V. K.; Pal, A.; Darokar, M. P.; Bhakuni, R. S.; Kumar, B. *In vitro* and *in vivo* synergistic interaction of substituted chalcone derivatives with norfloxacin against methicillin resistant *Staphylococcus aureus*. *RSC Adv.* **2015**, *5*, 5830–5845.
- (30) Lepri, S.; Buonerba, F.; Goracci, L.; Velilla, I.; Ruzziconi, R.; Schindler, B. D.; Seo, S. M.; Kaatz, G. W.; Cruciani, G. Indole based weapons to fight antibiotic resistance: a structure-activity relationship study. *J. Med. Chem.* **2016**, *59*, 867–891.
- (31) Brincat, J. P.; Broccatelli, F.; Sabatini, S.; Frosini, M.; Neri, A.; Kaatz, G. W.; Cruciani, G.; Carosati, E. Ligand promiscuity between the efflux pumps human p-glycoprotein and *S. aureus* NorA. *ACS Med. Chem. Lett.* **2012**, *3*, 248–251.
- (32) Manna, U.; Chutia, R.; Das, G. Entrapment of cyclic fluoride-water and sulfate-water-sulfate cluster within the self-assembled structure of linear meta-phenylenediamine based bis-urea receptors: positional isomeric effect. *Cryst. Growth Des.* **2016**, *16*, 2893–2903.
- (33) Manna, U.; Nayak, B.; Das, G. Dual guest [(Chloride)₃-DMSO] encapsulated cation-sealed neutral trimeric capsular assembly: meta-substituent directed halide and oxyanion binding discrepancy of isomeric neutral disubstituted bis-urea receptors. *Cryst. Growth Des.* **2016**, *16*, 7163–7174.
- (34) Jose, D. A.; Kumar, D. K.; Kar, P.; Verma, S.; Ghosh, A.; Ganguly, B.; Ghosh, H. N.; Das, A. Role of positional isomers on receptor–anion binding and evidence for resonance energy transfer. *Tetrahedron* **2007**, *63*, 12007–12014.
- (35) Casula, A.; Fornasier, M.; Montis, R.; Bettoschi, A.; Argent, S. P.; Blake, A. J.; Lippolis, V.; Marongiu, L.; Picci, G.; Tidey, J. P.; Caltagirone, C. Halogen-substituted ureas for anion binding: solid state and solution studies. *Supramol. Chem.* **2017**, *29*, 875–886.
- (36) Viveiros, M.; Martins, A.; Paixão, L.; Rodrigues, L.; Martins, M.; Couto, I.; Fahnrich, E.; Kern, W. V.; Amaral, L. Demonstration of intrinsic efflux activity of *Escherichia coli* K-12 AG100 by an automated ethidium bromide method. *Int. J. Antimicrob. Agents* **2008**, *31*, 458–462.
- (37) Giraud, E.; Cloeckert, A.; Kerboeuf, D.; Chaslus-Dancla, E. Evidence for active efflux as the primary mechanism of resistance to ciprofloxacin in *Salmonella enterica* serovar typhimurium. *Antimicrob. Agents Chemother.* **2000**, *44*, 1223–1228.
- (38) Thiagarajan, D.; Das, G.; Ramesh, A. Amphiphilic cargo-loaded nanomaterial enhances antibiotic uptake and perturbs efflux: effective synergy for mitigation of methicillin-resistant *Staphylococcus aureus*. *ChemMedChem* **2017**, *12*, 1125–1132.
- (39) Cartiera, M. S.; Ferreira, E. C.; Caputo, C.; Egan, M. E.; Caplan, M. J.; Saltzman, W. M. Partial correction of cystic fibrosis defects with PLGA nanoparticles encapsulating curcumin. *Mol. Pharmaceutics* **2010**, *7*, 86–93.
- (40) Saha, B.; Das, S.; Saikia, J.; Das, G. Preferential and enhanced adsorption of different dyes on iron oxide nanoparticles: a comparative study. *J. Phys. Chem. C* **2011**, *115*, 8024–8033.
- (41) Dey, P.; Das, G.; Ramesh, A. Interplay between supramolecular and coordination interactions in synthetic amphiphiles: Triggering metal starvation and anchorage onto MRSA cell surface. *Langmuir* **2020**, *36*, 2110–2119.

(42) Livak, K. J.; Schmittgen, T. D. Analysis of relative gene expression data using real-time quantitative PCR and the $2^{-\Delta\Delta CT}$ method. *Methods* **2001**, *25*, 402–408.

(43) Muñoz-Provencio, D.; Llopis, M.; Antolin, M.; de Torres, I.; Guarner, F.; Perez-Martínez, G.; Monedero, V. Adhesion properties of *Lactobacillus casei* strains to resected intestinal fragments and components of the extracellular matrix. *Arch. Microbiol.* **2009**, *191*, 153–161.

(44) Mukherjee, S.; Ramesh, A. Bacteriocin-producing strains of *Lactobacillus plantarum* inhibit adhesion of *Staphylococcus aureus* to extracellular matrix: Quantitative insight and implications in antibacterial therapy. *J. Med. Microbiol.* **2015**, *64*, 1514–1526.

(45) Singh, A. K.; Mukherjee, S.; Adhikari, M. D.; Ramesh, A. Fluorescence-based comparative evaluation of bactericidal potency and food application potential of antilisterial bacteriocin produced by lactic acid bacteria isolated from indigenous samples. *Probiotics Antimicrob Proteins* **2012**, *4*, 122–132.

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